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STRATEGIC ASSET ALLOCATION

Learning Outcomes

- explain the specification of asset classes in relation to asset allocation
- describe the principles of portfolio construction and the role of asset allocation in relation to the IPS

Traditionally, investors have distinguished cash, equities, bonds (government and corporate), and real estate as the major asset classes. In recent years, this list has been expanded with private equity, hedge funds, high-yield and emerging market bonds, and commodities. In addition, such assets as art and intellectual property rights may be considered asset classes for those investors prepared to take a more innovative approach and to accept some illiquidity. Combining such new asset classes as well as hedge funds and private equity under the header “alternative investments” has become accepted practice.

As the strategic asset allocation is built up by asset classes, the decision about how to define those asset classes is an important one. Defining the asset classes also determines the extent to which the investor controls the risk and return characteristics of the eventual investment portfolio. For example, separating bonds into government bonds and corporate bonds, and then further separating corporate bonds into investment grade and non-investment grade (high yield) and government bonds into domestic and foreign government bonds, creates four bond categories. For these categories, risk–return expectations can be expressed and correlations with other asset classes (and, in an asset–liability management context, with the liabilities) can be estimated. An investment manager who wants to explicitly consider the risk–return characteristics of those bond categories in the strategic asset allocation may choose to treat them as distinct asset classes. Similarly, in equities, some investors distinguish between emerging market and developed market equities, between domestic and international equities, or between large-cap and small-cap equities. In some regulatory environments for institutional investors, asset class definitions are mandatory, thereby forcing investment managers to articulate risk–return expectations (and apply risk management) on the asset classes specified. Conversely, a broader categorization of asset classes leaves the allocation between different categories of bonds and equities, for example, to managers responsible for these asset classes.

When defining asset classes, a number of criteria apply. Intuitively, an asset class should contain relatively homogeneous assets while providing diversification relative to other asset classes. In statistical terms, risk and return expectations should be similar, and paired correlations of assets should be relatively high within an asset class but should be lower versus assets in other asset classes. Also, the asset classes, while being mutually exclusive, should add up to a sufficient approximation of the relevant investable universe. Applying these criteria ensures that the strategic asset allocation process has considered all available investment alternatives.

EXAMPLE 9

Specifying Asset Classes

The strategic asset allocations of many institutional investors make a distinction between domestic equities and international equities or between developed market equities and emerging market equities. Often, equities are separated into different

between different equity asset classes and other asset classes. Specifically, these correlations are measured over the period from December 2000 through August 2018. In addition, the exhibit shows the annualized volatility of monthly returns.

Exhibit 5:

Asset Class Correlation Matrix

Correlations	US Equities	Emerging Markets	European Equities	Japanese Equities	US Small-Cap Equities	Commodities	E
US Equities	1.00	0.78	0.88	0.59	0.89	0.32	
Emerging Markets Equities	0.78	1.00	0.84	0.64	0.75	0.46	
European Equities	0.88	0.84	1.00	0.64	0.79	0.43	
Japanese Equities	0.59	0.64	0.64	1.00	0.57	0.32	
US Small-Cap Equities	0.89	0.75	0.79	0.57	1.00	0.32	
Commodities	0.32	0.46	0.43	0.32	0.32	1.00	
European Gov't. Bonds	0.08	0.21	0.16	0.24	0.09	0.13	
US Treasuries	–0.37	–0.24	–0.28	–0.18	–0.36	–0.18	
US Credits	0.19	0.34	0.29	0.29	0.19	0.12	
US High-Yield Credit	0.66	0.70	0.68	0.52	0.69	0.36	
Volatility	14.3%	21.6%	18.4%	15.6%	18.4%	22.3%	

Correlations and volatilities have been calculated using monthly returns from December 2000 through August 2018, unhedged, in USD.

Source: MSCI Bloomberg, S&P

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Based only on the information given, address the following:

1. Contrast the correlations between equity asset classes with the correlations between equity asset classes and US Treasuries.

   Hide Solution

   Solution:

   The matrix reveals very strong correlation between the equity asset classes. For example, the correlation between European equities and US equities is 0.88. The correlation of equities with bonds, however, is much lower. For example, US equities, emerging markets equities, European equities, and Japanese equities all have negative correlation with US government bonds (–0.37, –0.24 and –0.28, and –0.18, respectively). It is worth noting, however, that correlations can vary through time and the values shown may be specific to the sample period used.

2. The monthly returns of which equity asset class differ the most from US equities?

   Hide Solution

   Solution:

   Among equity asset classes as listed in the table, the correlation between US and Japanese equities is the lowest, at 0.59. By contrast, correlations

Using correlation as a metric, Example 9 tends to indicate that only emerging markets were well differentiated from European equities. So, why do investors still often subdivide equities? Apart from any regulatory reasons, one explanation might be that this decomposition into smaller asset classes corresponds to the way the asset allocation is structured in portfolios. Many investment managers have expertise exclusively in specific areas of the market, such as emerging market equities, US small-cap equity, or international investment-grade credit. Bringing the asset class definitions of the asset allocation in line with investment products actually available in the market may simplify matters from an organizational perspective.

The risk–return profile of the strategic asset allocation depends on the expected returns and risks of the individual asset classes, as well as the correlation between those asset classes. In general, adding assets classes with low correlation improves the risk–return trade-off (more return for similar risk). Typically, the strategic asset allocation for risk-averse investors will have a large weight in government bonds and cash, whereas those with more willingness and ability to take risk will have more of their assets in risky asset classes, such as equities and many types of alternative investments.

It is customary to represent asset classes using benchmarks and universes calculated by providers such as FTSE, MSCI, or Bloomberg. A negative screening or a **best-in-class** policy (discussed previously) limits the number of securities to choose from, potentially impacting the risk and expected return estimates for these asset classes. Some examples of exclusions may be controversial weaponry or tobacco companies, or investments in certain countries. When such exclusions apply, risk and return estimates based on non-traditional ("off-the-shelf") asset class benchmarks may not be applicable. Separate benchmark indices reflecting the exclusions may be available from the providers to mitigate this issue.

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ABP is the pension fund for the Dutch government sector employees. The fund offers teachers, police officers, members of the military, and other civil servants a defined benefit pension plan, aiming for a pension of 70% of the average career real income for employees. As of the first quarter of 2018, ABP had €405 billion under management. The strategic asset allocation as of this period is shown in Exhibit 6.

Exhibit 6:

Strategic Asset Allocation for ABP

Equity	
Equities, developed countries	27%
Equities, emerging markets	9%
Total equity	***36%***
Alternatives	
Real estate	10%
Private equity	5%
Hedge funds	4%
Commodities	5%
Infrastructure	3%
Total alternatives	***27%***
Fixed-income securities	
Government bonds	13%
Corporate bonds	13%

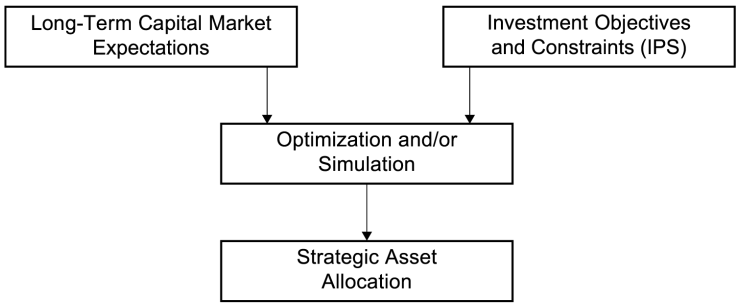
Total fixed income	***37%***
Total	100%

Source: ABP Quarterly Report Q1 2018

A strategic asset allocation results from combining the constraints and objectives articulated in the IPS and long-term capital market expectations regarding the asset classes. The strategic asset allocation or policy portfolio will subsequently be implemented into real portfolios. Exhibit 7 illustrates conceptually how investment objectives and constraints and long-term capital market expectations combine into a policy portfolio.

Exhibit 7:

Strategic Asset Allocation Process



▶ Expand / collapse Extended Description

In some frameworks used in practice, the asset allocation is an integral part of the investment policy statement. This presentation, however, keeps the asset allocation separate from the investment policy statement because clients' investment objectives and constraints qualitatively differ in nature from capital market expectations, thus requiring different types of analysis, different sources of information, and different review cycles.

The combination of investment objectives/constraints and capital market expectations theoretically occurs using optimization techniques. In this section, we apply mean–variance optimization to a sample set of investment objectives and constraints, using an investment universe with associated market expectations. We assume that investors choosing from a range of asset allocations with similar returns would prefer those with lower risk. Choosing from allocations with similar levels of risk, investors would prefer those with the highest return. Formally, investors' risk and return objectives can be described as a utility function, in which utility increases with higher expected returns and lower risk. This assumption could yield an expected utility equation such as that shown in Equation 1.[4]

$$U_p = E(R_p) - \lambda\sigma_p^2 \qquad 1$$

where

U_p = the investor's expected utility from the portfolio

$E(R_p)$ = the expected return of the portfolio

σ_p = the standard deviation of returns of the portfolio

λ = a measure of the investor's risk aversion

This utility function expresses a positive relationship between utility and expected portfolio return (i.e., higher expected return increases utility, all else equal) and a negative relationship between utility and volatility of portfolio return as measured by the variance of portfolio returns. The stronger the negative relationship, the greater the investor's risk aversion. The portfolio is understood to represent a particular asset allocation. The asset

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expected return that produces that level of utility: an indifference curve. An investor would attain equal utility from all risk–return combinations on that curve.

Capital market expectations, specified in asset classes' expected returns, standard deviations of return, and correlations, translate into an efficient frontier of portfolios. A multi-asset class portfolio's expected return is given by

$$E(R_p) = \sum_{i=1}^{n} w_i E(R_i) \quad (2)$$

where w_i equals the weight of asset class i in the portfolio, and its risk is given by

$$\sigma_p = \sqrt{\sum_{i=1}^{n} \sum_{j=1}^{n} w_{p,i} w_{p,j} \mathrm{Cov}(R_i, R_j)} \quad (3)$$

The covariance between the returns on asset classes i and j is given by the product of the correlation between the two asset classes and their standard deviations of return:

$$\mathrm{Cov}(R_i, R_j) = \rho_{i,j} \sigma_i \sigma_j \quad (4)$$

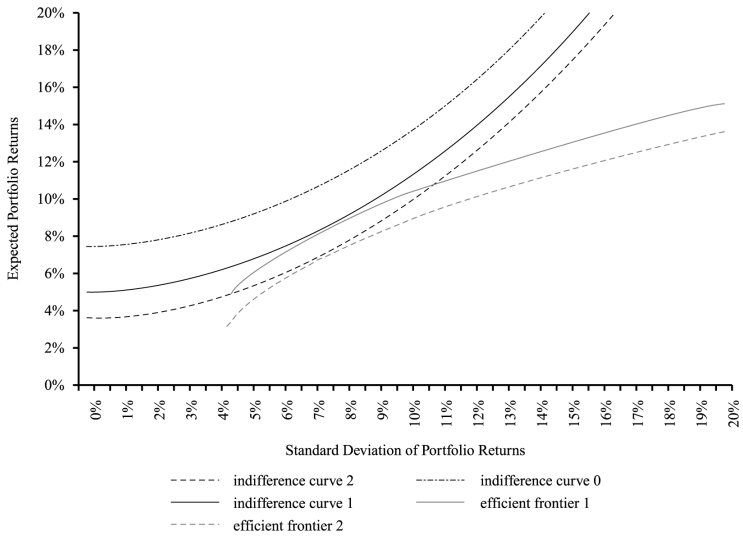
where

$\mathrm{Cov}(R_i, R_j)$ = the covariance between the return of asset classes i and j

$\rho_{i,j}$ = the correlation between the returns of asset classes i and j

The resulting portfolios can be represented as a scatter of dots in a chart depicting their risk and expected return. Because a portfolio's risk is a positive function of the risk of its assets and the correlations among them, a portfolio consisting of risky assets with low correlation has lower risk than one with similarly risky assets with high correlation. It is therefore possible to construct different portfolios with equal expected returns but with different levels of risk. The line that connects those portfolios with the minimal risk for each level of expected return (above that of the **minimum-variance portfolio**—the portfolio with the minimum variance for each given level of expected return) is the efficient frontier. Clearly, the efficient frontier will move "upward" as more low-correlation assets with sufficient expected return are added to the mix because it lowers the risk in the portfolios for equal expected returns. Similarly, when return expectations increase for asset classes while volatility and correlation assumptions remain unchanged, the efficient frontier will move upward because each portfolio is able to generate higher returns for the same level of risk.

Both the efficient frontier and a range of indifference curves can be plotted in the risk–return space. In Exhibit 8, the dark-colored curves that are concave from below represent efficient frontiers associated with different assumed expected returns. The lighter-colored curves are indifference curves. The point where the efficient frontier intersects with the indifference curve with the highest utility attainable (i.e., the point of tangency) represents the optimal asset allocation for the client/investor. In Exhibit 8, efficient frontier 1 has a point of tangency with indifference curve 1. Higher levels of utility, such as those associated with indifference curve 0, can apparently not be reached with the assets underlying the efficient frontier. It is clear that when capital market expectations change, this change moves the efficient frontier away from its original location. In the chart, this movement is illustrated by efficient frontier 2, which incorporates different capital market expectations. This new efficient frontier has a point of tangency with indifference curve 2, which is associated with a lower level of expected utility. Because the point of tangency represents the strategic asset allocation, it implies the asset allocation should be adjusted. Similarly, should investment objectives or constraints change, the indifference curves will change their shape and location. This change will again move the point of tangency, and hence change the asset allocation.



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This framework describes how investor objectives and capital market expectations should theoretically be reconciled. It will, however, not be the exact procedure that in practice will be followed. First, an IPS does not necessarily translate the client's investment objectives and constraint into a utility function. Rather, an IPS gives threshold levels for risk and expected return, combined with a number of additional constraints that cannot be captured in this model. Second, the model illustrated is a single-period model, whereas in practice, the constraints from the IPS will make it more appropriate to use multi-period models. Multi-period problems can be more effectively addressed using simulation.

EXAMPLE 10

Approaching a SAA for a Private Investor

1. Rainer Gottschalk recently sold his local home construction company in the south of Germany to a large homebuilder with a nationwide reach. Upon selling his company, he accepted a job as regional manager for that nationwide homebuilder. Gottschalk is now considering his and his family's financial future. He looks forward to his new job—he likes his new role, and the position provides him with income to fulfill his family's short-term and medium-term liquidity needs. Gottschalk feels strongly that he should not invest the proceeds of the sale of his company in real estate because his income already depends on the state of the real estate market. Also, reflecting family values, he feels strongly that his savings should not support the tobacco industry. He therefore wants his equity allocation to exclude any stocks of tobacco product manufacturers or retailers. Gottschalk consults a financial adviser from his bank about how to invest his money to retire in good wealth in 20 years.

   The IPS developed by his adviser suggests a return objective of 5%, with a standard deviation of 10%. The bank's asset management division provides Gottschalk and his adviser with the following data (Exhibit 9, Panel 1) on market expectations. The adviser estimates that excluding the tobacco industry from the investment universe affects expected equity returns of European equities by –0.2% and annual standard deviation by +0.1%. The impact on emerging market equities, and on the correlation structure, was considered negligible. Gottschalk accepts the results of these calculations as shown in Exhibit 9, Panel 2.

Exhibit 9:

	Expected Return	Standard Deviation	European Equities	Emerging Mkt Equities	European Govt Bonds
Panel 1					
European equities	6.0%	15.0%	1.00	0.78	–0.08
Emerging market equities	8.0%	20.1%	0.78	1.00	–0.07
European government bonds	2.0%	7.8%	–0.08	–0.07	1.00
Panel 2					
European equities	5.8%	15.1%	1.00	0.78	–0.08
Emerging market equities	8.0%	20.1%	0.78	1.00	–0.07
European government bonds	2.00%	7.8%	–0.08	–0.07	1.00

Standard deviation and correlation calculated over the period March 1999–August 2018. All data in unhedged euros.
Sources: MSCI, Bloomberg

To illustrate the possibilities, the adviser presents Gottschalk with the following plot (Exhibit 10), in which the points forming the shaded curve outline the risk–return characteristics of the portfolios that can be constructed out of the three asset classes. An imaginary line linking the points with the lowest standard deviation for each attainable level of return would be the efficient frontier. The two straight lines show the risk and return objectives. Gottschalk should aim for portfolios that offer an expected return of at least 6% (the straight horizontal line or above) and a standard deviation of return of 12% or lower (the straight vertical line to the left).

Exhibit 10: Efficient Frontier

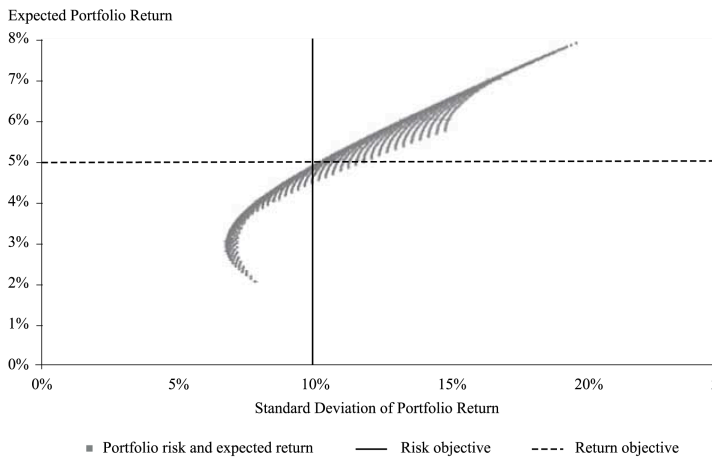

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Exhibit 10 shows that no portfolio satisfies the two objectives (return of 5% and standard deviation of 10%) exactly, because the highest expected return that can be attained at a maximum volatility of 10% is 4.9%. This difference, Gottschalk and the adviser agree, is acceptable. The portfolio that would correspond with this expected return consists of 16% European stocks, 38% emerging market equities, and 46% government bonds.

Discussion

What do you want to discuss... (required)

RR The Utility formula above in Equation 1 ($Up=E(Rp)-\lambda\sigma 2p$) does not have 1/2 but the same Utility formula in Portfolio Risk and Return Part 1 is without 1/2 i.e (U = E(r) - 1/2 Aσ^2. Any reason why so?

Created 6 months ago by Rahul Ramchandra Naik

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